

Digital-Economy Development and the Skill Composition of Chinese Employment

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This work is the sole responsibility of the candidate.

This version (July 2026) corrects data errors and language in the dissertation submitted in April 2025; results are unchanged in direction.

The replication data and Stata code for this dissertation are openly available at https://github.com/airuimeng1/UG_thesis_code_repository.

Abstract

This paper examines how the development of China's digital economy is associated with the skill composition of employment, using panel data on 31 provincial-level administrative units (excluding Hong Kong, Macao, and Taiwan) from 2013 to 2022. Employment is divided into low-, medium-, and high-skilled shares by educational attainment. A three-dimensional index of digital infrastructure, digital industry development, and the digital economy environment is constructed with the entropy weight method (EWM) and, alternatively, principal component analysis (PCA). Two-way fixed-effects models with a centered quadratic term are estimated and checked with alternative indices, wild cluster bootstrap, and instrumental variables. The digital economy has an inverted U-shaped association with low-skilled employment: the positive association weakens as digital development rises, but the turning point lies at the upper edge of the data, so later-stage job substitution is a projection. It has a U-shaped association with high-skilled employment, but the upward segment, observed only in Guangdong, is not statistically confirmed. No significant association is found for medium-skilled employment, possibly because automation substitution and the creation of new jobs offset each other. The positive association with low-skilled employment is significantly weaker in coastal than in inland provinces. The nonlinear results are robust to the composition of the index and to equal weighting but not to conventional entropy weights, and the high-skilled curvature survives neither excluding Guangdong nor using log-ratio outcomes; the instrumental-variable estimates do not support a causal interpretation. The findings suggest strengthening digital skills training and promoting regionally coordinated development to facilitate inclusive growth.

Keywords: human capital, employment structure, digital economy

Digital-Economy Development and the Skill Composition of Chinese Employment

The digital economy represents a transformative economic activity driven by advancements in digital technologies and serves as a catalyst for economic modernization by refining resource allocation, enabling data-informed decision-making, and balancing equity with productivity. Distinct from traditional economic models, it is characterized by high automation and data-driven processes, which have a significant impact on the Chinese labor market (Xia & Pei, 2021). This not only fosters the creation of innovative professions, but also raises competency standards for workers.

Against the background of persistent global headwinds marked by sluggish growth, rising geopolitical trade tensions, and domestic economic cooling, the digital economy, underpinned by breakthroughs in information and communication technologies (ICT), is fundamentally transforming employment structure. By accelerating structural adjustment across sectors, it disrupts conventional labor dynamics while fostering innovation-driven industrial transitions. The pervasive integration of digital tools into production networks, supply chains, and consumer ecosystems has not only deepened the digital economy's societal footprint but also redefined the interplay between technology and human capital.

Therefore, an examination of the digital economy's impact on employment structure is imperative to reconcile its possible disruptive and creative forces. By devising evidence-based interventions, policymakers can harness its potential to expand job availability, elevate labor conditions, and realign occupational frameworks, ensuring inclusive growth in the digital era.

Based on data from 31 provincial-level administrative units in China (excluding the Hong Kong and Macao Special Administrative Regions and Taiwan Province), covering the period from 2013 to 2022, this paper uses panel data to study the changes in the employment structure of human capital associated with the development of China's digital economy.

The research focuses on one core aspect of the employment structure: distribution of human capital. Empirical analysis shows that the digital economy is significantly and nonlinearly associated with the skill composition of employment, and that this association differs between

inland and coastal provinces. Compared with existing studies, this paper achieves a dual expansion in the temporal and spatial dimensions: spatially covering all 31 provincial-level administrative units and extending the time series to the relatively mature development period of the digital economy, revealing in depth the new characteristics of the evolution of the employment structure under the emerging economic form. The innovation lies in the construction of a nonlinear association analysis framework, clarifying the differentiated patterns of the digital economy's association with the employment structure of human capital. This not only provides theoretical support for judging the employment trend in the digital economy era, but also offers a decision-making basis with both theoretical depth and practical value for the government to formulate industrial optimization policies and improve the employment promotion mechanism.

Literature Review

The digital economy's transformative impact on global employment presents a complex duality of productivity enhancement and structural disruption.

Non-Standardization of Employment Patterns and Reconstruction of Labor Relations

Digital technologies have catalyzed the emergence of flexible employment models, leading to the deconstruction of traditional employer-employee relationships. Ignatieva et al. (2023) highlight the surge of non-standard employment forms, such as remote work and flexible hours, which reflect a “de-socialization” of labor relations. Corporate organizational structures have shifted from vertical hierarchies to horizontal collaboration, with team-based and project-oriented work becoming central. While this transformation reduces labor costs for enterprises, it simultaneously exacerbates employment instability for workers. C. Wu (2021) further corroborates this trend, demonstrating the rise of the gig economy and platform-based employment in China, where the proportion of flexible employment increased significantly between 2013 and 2017. However, this shift is accompanied by inadequate social security coverage and heightened occupational risks. Taken together, these findings underscore that while the digital economy enhances labor market flexibility, there is an urgent need to establish institutional frameworks compatible with these emerging labor relations.

Structural Tensions Between Job Creation and Substitution Effects

The impact of the digital economy on aggregate employment exhibits marked heterogeneity across industries and regions. Based on classical economic theory, the Petty–Clark theorem provides a foundational framework for understanding labor migration patterns, positing that income differentials across industries drive workforce reallocation toward sectors with higher productivity and value-added returns. Contemporary empirical studies confirm this theorem’s enduring relevance in the digital era, particularly through observable employment shifts from secondary to tertiary industries. Y. Zhao and Said (2023) specifically quantify this transition, revealing significant employment contraction in the manufacturing and construction sectors alongside service sector expansion, a structural realignment mirroring economic complex movement, which is also quantified by Lu and Zhou (2023) via a spatial Durbin model. Notably, Lu and Zhou (2023) find job creation effects are stronger in small and medium-sized cities than in large cities. This “3-1-2” industrial restructuring highlights the digital economy’s role in driving cross-sector labor reallocation, yet low-skilled workers face heightened vulnerability to technological displacement (C. Wu, 2021).

In addition, Xia and Pei (2021), using China’s pandemic period as an observational window, reveal that digital technologies generated substantial employment opportunities through new formats like e-commerce and remote work, yet simultaneously disrupted the traditional manufacturing and brick-and-mortar retail sectors, resulting in structural unemployment.

Skill Demand Upgrading and Digital Divide Challenges

The digital transformation of employment structures imposes higher skill requirements on workers. Ignatieva et al. (2023) identify advanced educational attainment and digital literacy as critical thresholds for emerging occupations, with women’s participation increasing in knowledge-intensive sectors. However, workers in traditional industries confront the risk of skill obsolescence. Rural-urban disparities in digital infrastructure exacerbate the “information divide” (Xia & Pei, 2021). Lu and Zhou (2023) emphasize that innovation and entrepreneurship levels serve as key mediators in alleviating skill mismatches, necessitating enhanced industry-academia

collaboration and vocational training to improve human capital adaptability. These findings highlight the urgency of reforms in skill supply systems and the establishment of lifelong learning mechanisms.

Regional Heterogeneity and Differentiated Policy Pathways

The employment effects of the digital economy demonstrate significant spatial divergence. Lu and Zhou's (2023) empirical analysis reveals that in China's eastern regions, characterized by high economic agglomeration, the digital economy directly stimulates service-sector employment growth. In contrast, central and western regions rely on indirect spillovers from innovation and entrepreneurship, requiring policies focused on digital infrastructure and industrial incubation. Xia and Pei (2021) advocate targeted interventions in underdeveloped areas, such as digital skills training programs ("customized training") and hours-based social security contribution systems, while C. Wu (2021) proposes harmonized urban-rural household registration policies to facilitate labor mobility. These region-specific strategies provide actionable insights for addressing the uneven distribution of "digital dividends."

Methodology

Considering the established theoretical framework in the fields of digital economics and labor market studies, this study utilizes a two-way fixed-effects panel data model, given the multi-year span of the data and their coverage of 31 Chinese provinces. This econometric approach is chosen to systematically examine the dynamic associations between the development of the digital economy and the structure of employment in China, accounting for both time-invariant individual heterogeneity and time-specific macroeconomic shocks. The literature reviewed above suggests that the digital economy both creates and displaces jobs, and that the balance between these forces may shift as digital development deepens. To allow for such a nonlinear relationship, the model includes the square of the digital economy index. The model specification is presented as follows:

$$Y_{it} = \beta_0 + \beta_1 \widehat{digital}_{it} + \beta_2 (\widehat{digital}_{it})^2 + \sum_j \theta_j X_{itj} + \mu_i + \rho_t + \epsilon_{it} \quad (1)$$

Note: The subscript i represents different provinces or municipalities, the subscript t represents the year (2013–2022), Y_{it} represents the employment structure of human capital (*low_skill_pct*, *mid_skill_pct*, *high_skill_pct*) in province i in the t -th year, $\widetilde{digital}_{it}$ represents the development level of the digital economy in province i in the t -th year, centered at its sample mean of 0.224, X_{itj} represents other control variables, μ_i is the individual effect, ρ_t is the time effect, and ϵ_{it} is the random error term.

(Note: The italicized words in parentheses are the variable names.)

Centering the index before squaring leaves β_2 and the fitted relationship unchanged but reduces the mechanical correlation between the index and its square (from $r = .958$ to $.697$), and β_1 becomes the marginal association at the sample mean. A negative β_2 indicates an inverted U-shaped relationship and a positive β_2 a U-shaped one, with the turning point located at $\overline{digital} - \beta_1/(2\beta_2)$. Following Lind and Mehlum (2010), a turning point is accepted only if the slope changes sign significantly within the observed range of the index. Standard errors are clustered by province, which makes them robust to heteroskedasticity and to serial correlation within provinces; because the sample contains only 31 provinces, the key coefficients are also re-tested with the wild cluster bootstrap (Roodman et al., 2019).

In panel data analysis, model specification requires rigorous verification of data-model consistency, particularly between fixed-effects and random-effects specifications. This research uses Hausman tests for model selection. As demonstrated in Table 1, the Hausman statistics reject, at the 1% level, the null hypothesis that the fixed- and random-effects estimates do not differ systematically for medium- and high-skilled employment, but not for low-skilled employment ($\chi^2(7) = 2.52, p = .926$). Because the difference between the two covariance matrices was not positive definite in any of the three tests, the statistics are treated as indicative rather than decisive, and the fixed-effects specification is used for all three outcomes, since the fixed-effects estimator remains consistent whether or not the random-effects assumption holds.

Table 1*Hausman Test*

Dependent variable	χ^2 statistic	df	p value	Result
low_skill_pct	2.52	7	.926	Do not reject
mid_skill_pct	43.60	7	< .001	Reject
high_skill_pct	259.10	7	< .001	Reject

Note. low_skill_pct, mid_skill_pct, and high_skill_pct denote the employment shares of low-, medium-, and high-skilled workers. The test compares fixed- and random-effects estimates of the quadratic specification without year fixed effects. In all three tests the difference between the two covariance matrices was not positive definite, so the statistics should be read as indicative.

Data and Variables**Variable Measurement***Explained Variables*

This paper uses the employment structure of human capital as the core explained variable. It is divided into three levels based on educational attainment: the group with an educational attainment of primary school or below is defined as low human capital (*low_skill_pct*), the group with an educational attainment from junior high school to junior college is classified as intermediate human capital (*mid_skill_pct*), and the group with a bachelor's degree or above is regarded as high human capital (*high_skill_pct*).

The shares are calculated from the educational composition of employed persons reported in the China Labor Statistical Yearbook. Two features of this measure should be kept in mind. First, it describes the educational attainment of the employed population rather than the skill requirements of jobs. This is a supply-side characteristic that also reflects the expansion of higher education and generational change. The regressions therefore capture associations with the educational composition of employment, and interpreting them as shifts in skill demand requires additional assumptions. Second, in 2015–2018 the three shares do not sum to 100: in all 124 province–years of those four years they sum to between 87.2% and 98.4% (about 94% on

average), which indicates that at least one educational category is missing from the yearbook data for those years. Year fixed effects absorb the common part of this shortfall, and its influence on the estimates is examined in the Robustness Tests section.

Core Explanatory Variables

The core explanatory variable of this study is the development level of the digital economy (*digital_e*, *digital_p*), and the entropy weight method (EWM) and principal component analysis (PCA) are adopted for comprehensive measurement.

The construction of the indicator system builds upon the methodologically rigorous research framework proposed by He et al. (2023), systematically incorporating and synthesizing three core dimensions critical to digital economy analysis: the supporting capacity of digital infrastructure, which underpins the digital economy's foundational technological and infrastructural prerequisites; the growth capacity of the digital industry, reflecting the dynamic expansion and innovation potential of digital-related sectors; and the development environment of the digital ecosystem, encompassing institutional, regulatory, and collaborative conditions that foster digital economic activities. This hierarchical system is composed of 3 first-level indicators, 7 carefully selected second-level indicators, and 22 third-level indicators (see the detailed classification in Table 2), which together form a comprehensive composite index through multi-dimensional data aggregation. This approach ensures that indicators of infrastructure, industrial development, and the wider digital environment are systematically integrated to capture the complex landscape of digital economic development.

Notably, the study employs a rigorous cross-checking approach using two distinct measurement methodologies: the entropy weight method (EWM) and principal component analysis (PCA). By applying these complementary techniques, the research not only safeguards the objectivity and reliability of index weighting schemes but also enhances the comprehensive index's explanatory capacity across different analytical perspectives. This dual-method validation strategy mitigates potential biases inherent in single-weighting approaches, thereby enabling a more accurate and careful measurement of the digital economy's dynamic developmental

Table 2*Digital Economy Measurement Indicators and Their EWM Weights*

Primary category	Secondary category	Third-level indicator	Direction	Weight
Digital infrastructure	Hardware facilities	Long-distance optical cable length (km)	+	0.0499
		Internet broadband access ports (10,000 units)	+	0.0484
		Mobile phone base stations (10,000 units)	+	0.0484
	Software facilities	Number of internet domain names (10,000 units)	+	0.0425
		Number of IPv4 addresses (10,000 units)	+	0.0418
		Number of internet websites (10,000 units)	+	0.0431
Digital industry development	Digital industrialization	Software business revenue (100 million yuan)	+	0.0377
		Telecommunications business revenue (100 million yuan)	+	0.0433
	Industrial digitalization	Number of websites per 100 enterprises (units)	+	0.0520
		Percentage of enterprises with e-commerce activities (%)	+	0.0508
		E-commerce sales revenue (100 million yuan)	+	0.0417
Digital economy environment	Application environment	Number of computers per 100 people (units)	+	0.0498
		Mobile internet users (10,000 households)	+	0.0487
		Mobile phone subscribers (year-end, 10,000 households)	+	0.0489
		Digital TV subscribers (10,000 households)	+	0.0489
		Digital Inclusive Finance Index	+	0.0508
		Digital–real economy integration index	+	0.0379
	Talent environment	Percentage of IT-related employment (%)	+	0.0444
		Number of bachelor's degree graduates (persons)	+	0.0495
	Innovation environment	Full-time equivalent (FTE) of R&D personnel (person-years)	+	0.0444
		Number of R&D institutions of industrial enterprises above designated size (units)	+	0.0357
		Region-specific patent grants (units)	+	0.0413

Note. EWM = entropy weight method; + = positive indicator (all 22 indicators are positive in direction). The weights are those produced by the implementation described in the Entropy Weight Method section, which uses $k = 1/\ln(10)$.

characteristics over time and across regions. In practice, however, the two indices are very highly correlated ($r = .985$), so the PCA-based index serves as a consistency check on the weighting scheme rather than as an independent measure.

Control Variables

To effectively control for the influence of other potential factors on the employment structure by education level, this study introduces the following five control variables: (1) The level of industrialization (*industry*) is calculated as industrial value added divided by GDP. (2) The intensity of government intervention (*gov_intervention*) is reflected by the ratio of fiscal expenditure to GDP, which indicates the intensity of policy intervention. (3) The level of innovation input (*rd_intensity*) is measured by the ratio of research and development expenditure to GDP, which indicates the intensity of technological innovation. (4) Information infrastructure (*digitalization*), the degree of informatization development, is measured by the ratio of the total volume of postal and telecommunications services to GDP. (5) The human capital reserve (*human_capital*) is quantified by the ratio of the number of students in higher education to the permanent resident population. Two controls overlap conceptually with the digital economy index: *human_capital* is related to the number of bachelor's degree graduates and *rd_intensity* to the full-time equivalent of R&D personnel, both of which are index indicators (Table 2). The Robustness Tests section therefore re-estimates the model with these and the other labor-market indicators removed from the index.

Data Tests

Before the regression, the Pearson correlation coefficient matrix was computed (shown in Table 3). The results indicate that there is a significant relationship between the core explanatory variable, the level of digital economic development, and the explained variable, the employment skill structure: the level of digital economic development is significantly correlated with the skill structure of employment.

However, considering that the correlation coefficient matrix only measures the relationship between two variables and does not exclude the influence of control variables and

Table 3*Correlation Matrix of the Variables*

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
1. low_skill_pct	1											
2. mid_skill_pct	-.742***	1										
3. high_skill_pct	-.495***	-.155***	1									
4. digital_e	-.367***	.159***	.348***	1								
5. digital_e2	-.308***	.133**	.302***	.958***	1							
6. digital_p	-.398***	.111*	.474***	.985***	.958***	1						
7. digital_p2	-.383***	.102*	.466***	.976***	.973***	.997***	1					
8. human_capital	-.530***	.309***	.444***	.159***	.072	.196***	.172***	1				
9. rd_intensity	-.587***	.065	.812***	.626***	.553***	.691***	.676***	.530***	1			
10. digitalization	.074	-.025	-.008	.046	.051	.056	.054	-.065	-.118**	1		
11. gov_intervention	.745***	-.710***	-.163***	-.465***	-.356***	-.435***	-.406***	-.453***	-.445***	.148***	1	
12. industry	-.326***	.567***	-.292***	.108*	.089	.027	.020	.097*	.007	-.229***	-.572***	1

Note. * $p < .10$. ** $p < .05$. *** $p < .01$.

other potential factors (such as time effects and individual effects), the results are only indicative. The specific relationship still needs to be determined through further regression analysis.

To address multicollinearity in macroeconomic data, this study employs variance inflation factor (VIF) diagnostics. VIF quantifies collinearity severity by comparing regression variances under collinear versus ideal orthogonal conditions. Established thresholds are as follows: $VIF < 10$ suggests negligible collinearity; $10 \leq VIF < 100$ signals moderate collinearity; $VIF \geq 100$ denotes severe collinearity. Because the benchmark model contains both the index and its square, which are almost perfectly correlated in raw form ($r = .958$), the index is centered at its sample mean before squaring. As shown in Table 4 below, all VIF values are then below 10 (maximum 3.99, mean 2.36), and they remain below 10 when the year dummies are added (maximum 6.55 for the non-year regressors). Without centering, the VIFs of the index and its square would be 18.02 and 14.87, which reflects the mechanical correlation between a variable and its square rather than substantive multicollinearity among the regressors.

Table 4

VIF Diagnostics for Multicollinearity

Variable	VIF	1/VIF
digital_e (centered)	3.99	0.251
gov_intervention	2.85	0.351
rd_intensity	2.59	0.386
digital_e ² (centered)	2.36	0.425
industry	1.82	0.550
human_capital	1.80	0.557
digitalization	1.12	0.892
Mean VIF	2.36	

Note. VIF = variance inflation factor. VIFs are computed from a pooled OLS regression of low_skill_pct on the regressors of the benchmark quadratic specification, with digital_e centered at its sample mean before squaring. Without centering, the VIFs of digital_e and its square are 18.02 and 14.87.

Data Processing

Entropy Weight Method (EWM)

To scientifically measure the development level of the digital economy, it is necessary to consider both the systematic nature of the indicator system and the objectivity of weight assignment. To this end, this paper adopts the data-driven EWM for objective weighting.

This method quantifies the degree of dispersion of the indicators through the principle of information entropy, effectively avoids human interference, and ensures that the weight assignment is more in line with the characteristics of data distribution. The research period spans 2013 to 2022, a temporal framework meticulously chosen to align with critical milestones in China's digital economy trajectory. This time window closely corresponds to pivotal developmental phases, including the large-scale commercialization of 4G technology (which catalyzed the mobile internet boom) and the subsequent rollout of 5G infrastructure (marking the advent of high-speed, low-latency connectivity), as well as the national implementation of the Big Data Strategy in 2015. By encompassing these transformative periods, the study effectively captures the dynamic evolutionary associations of emerging digital technologies (such as mobile payments, e-commerce proliferation, and industrial digitalization) with labor market structures. This longitudinal design enables the identification of both short-term shocks and long-term trend changes associated with technological diffusion and policy interventions.

In the data processing stage, addressing the dimensional disparities among the 22 third-level indicators listed in Table 2 necessitated a rigorous standardization protocol. The research follows the entropy method framework proposed by He et al. (2023), whose steps are set out below, with one deviation in the normalizing constant described after Equation (6). First, the original data undergo range-standardization to eliminate unit-of-measurement discrepancies, ensuring that indicators with varying scales are converted into dimensionless metrics. This step mitigates the bias caused by dimensional heterogeneity, allowing for meaningful cross-indicator comparisons. Second, information entropy values are calculated for each indicator to objectively determine weight coefficients, a process that leverages data-driven statistical properties to reflect

the relative importance of each indicator based on its information richness. Finally, through entropy-weighted aggregation, the methodology achieves two key objectives: transforming multi-dimensional data into a unified comparable space and synthesizing a comprehensive index that encapsulates the complex interplay of digital economy dimensions. This systematic approach not only maintains the technical rigor of quantitative analysis but also ensures the resultant indices are theoretically grounded and empirically robust for panel data modeling.

The specific process of the entropy weight method is as follows:

1. Standardize the original data for all indicators (using normalization methods).

(Since all selected indicators are positive in direction, the standardization formula for negative indicators was omitted. This process removes dimensional discrepancies and improves data comparability.)

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (2)$$

Note: x'_{ij} represents the value of the i -th sample for the j -th indicator, and $\min(x_j)$ and $\max(x_j)$ are the minimum and maximum of indicator j over all 310 province-years.

2. Calculate the entropy of each indicator.

$$E_j = -k \sum_{i=1}^n p_{ij} \ln(p_{ij}) \quad (3)$$

Note: $p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^n x'_{ij}}$, $k = \frac{1}{\ln(n)} > 0$, $E_j \geq 0$, where $n = 310$ is the number of province-years.

3. Calculate the redundancy rate and weight.

$$d_j = 1 - E_j \quad (4)$$

$$w_j = \frac{d_j}{\sum_j d_j} \quad (5)$$

4. Calculate the score (i.e., the Digital Economy Comprehensive Development Index).

$$s_i = \sum_{j=1}^{22} w_j x'_{ij} \quad (6)$$

Note: $s_i \in [0, 1]$; a larger s_i means a higher level of digital economy development.

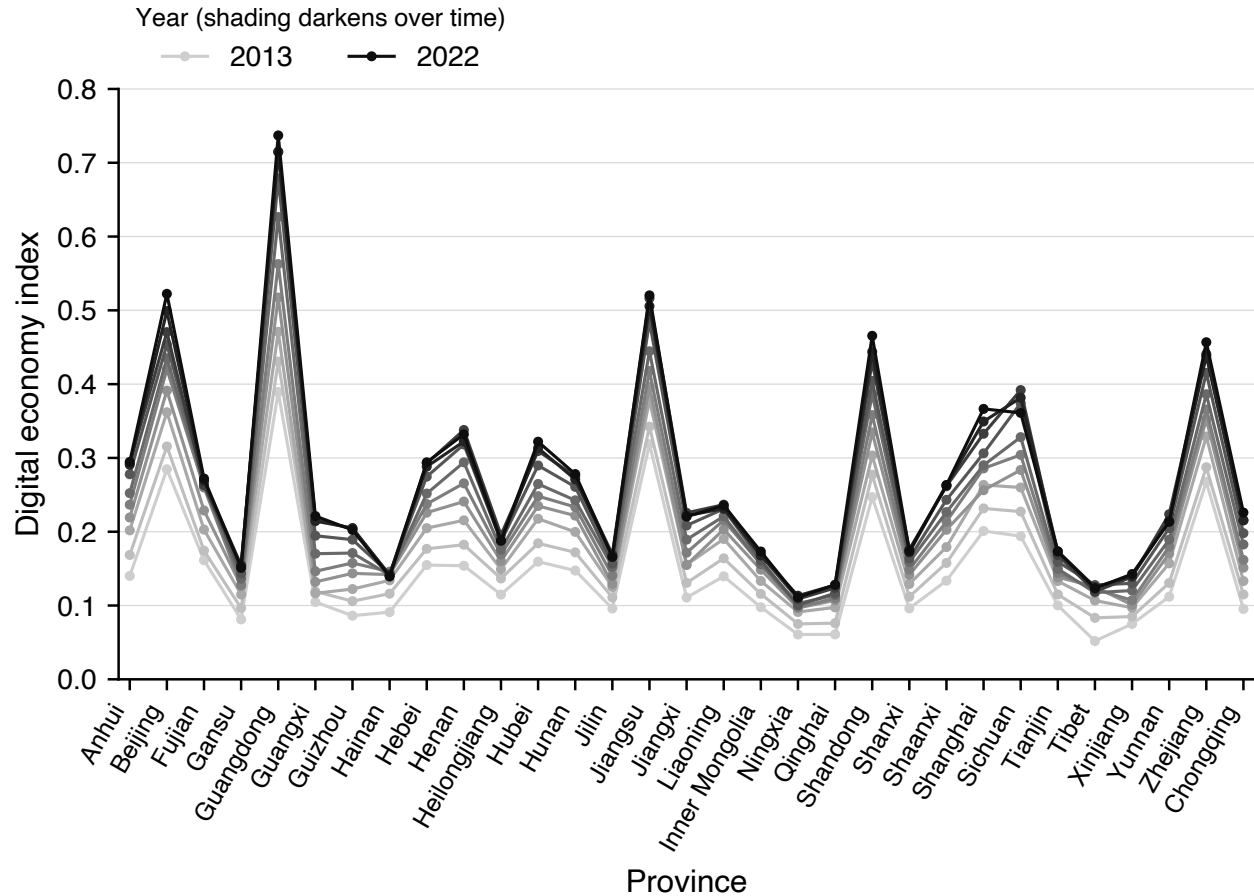
In the implementation used for this paper, the normalizing constant was set to $k = 1/\ln(T)$, where $T = 10$ is the number of years per province, rather than to $k = 1/\ln(n)$ with $n = 310$ as in Equation (3). Because the entropy is summed over all 310 province–years, every E_j then exceeds one (2.01–2.47) and every d_j is negative, and the resulting weights (ratios of negative numbers) are close to uniform, ranging from 0.036 to 0.052 (Table 2), and rank the indicators in exactly the reverse order of conventional entropy weights, so that the indicator with the lowest entropy receives the smallest weight. The composite index used below is therefore almost identical to an equal-weighted index ($r = .9996$). With the conventional constant $k = 1/\ln(n)$, the weights would range from 0.005 to 0.105 and would concentrate on a few indicators with high dispersion. The Robustness Tests section reports the results under both alternatives.

Figures 1 and 2 illustrate China’s digital economy composite index as calculated by EWM.

The empirical results reveal a notable upward trend in the comprehensive digital economy index, which rose from 0.146 in 2013 to 0.273 in 2022, translating to a compound annual growth rate (CAGR) of approximately 7.22%. This sustained growth reflects the cumulative effects of technological innovation, policy-driven digital transformation, and expanding market adoption of digital services across China. While all provincial-level administrative units experienced growth in digital economy development, the overall national foundation for digitalization remains moderately weak, accompanied by pronounced regional disparities that warrant detailed examination. Specifically, the Western region emerged as the fastest-growing area with a CAGR of 8.06%, driven by strategic national initiatives such as the “Belt and Road” digital infrastructure projects and local investments in cloud computing and big data hubs. The Central region followed closely with a CAGR of 8.05%, fueled by industrial digitalization upgrades and the rise of e-commerce clusters in provinces like Henan and Hubei (Luo, 2017). The Eastern region, despite

Figure 1

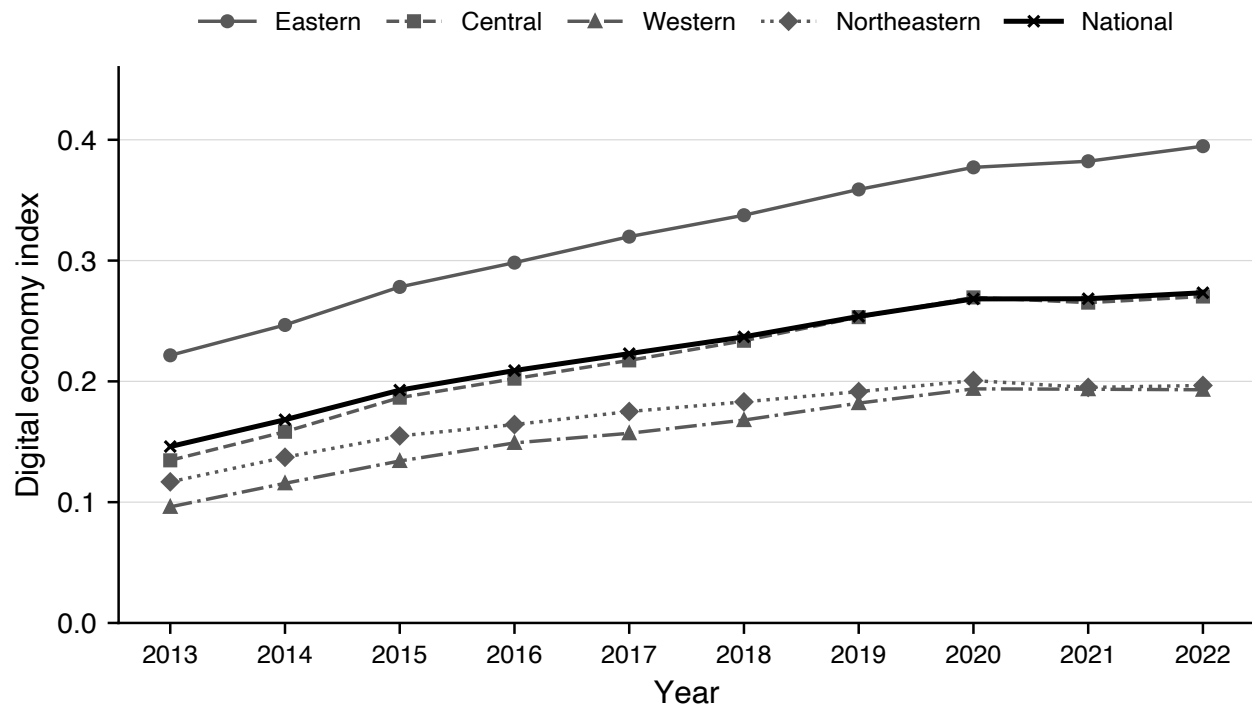
Digital Economy Comprehensive Development Index of 31 Provincial-Level Administrative Units (EWM)



Note. EWM = entropy weight method. Line shading darkens from 2013 to 2022.

its leading position in digital economy maturity, recorded a relatively low CAGR of 6.63%, likely due to its already high base from early digital adoption in the manufacturing and services sectors. The Northeastern region trailed with the lowest CAGR of 5.96%, constrained by challenges such as delays in restructuring traditional industries and talent outflows, which hindered the pace of digital innovation.

Averaged over 2013–2022, the national digital economy index stood at 0.224, with significant variations across regions. The Eastern region maintained its dominant position with an average index value of 0.322 (43.6% higher than the national average), benefiting from

Figure 2*Regional Digital Economy Development Index (EWM)*

Note. EWM = entropy weight method.

agglomeration effects in tech hubs like the Yangtze River Delta and Pearl River Delta, where advanced digital industries (e.g., artificial intelligence, fintech) and robust venture capital ecosystems thrive. The Central region ranked second with an index of 0.219, demonstrating substantial progress in digital infrastructure expansion but still lagging the East, whose average index is 46.8% higher, primarily due to disparities in high-tech talent concentration and R&D investment intensity. The Northeastern region reported an index of 0.171, while the Western region recorded 0.158—both falling below the national average. These figures highlight that although the Western and Central regions have achieved higher growth rates over 2013–2022, their lower starting bases and ongoing challenges in digital talent retention and institutional innovation mean they still face considerable hurdles. The Northeastern region’s performance underscores the need for targeted policies to revitalize its digital economy through state-owned enterprise digitalization reforms and regional collaboration initiatives. Collectively, while the

Eastern region's entrenched advantages in digital economy development are evident, the varying growth dynamics across regions emphasize the necessity of sustained investments in infrastructure, human capital, and policy coordination to foster more balanced national digital transformation.

Principal Component Analysis (PCA)

This paper employs the principal component analysis (PCA) methodology proposed by Tong (2002) to perform a weighted integration of the 22 observed variables. To mitigate potential biases caused by dimensional heterogeneity, all raw indicators underwent standardization preprocessing. Both Bartlett's sphericity test and the Kaiser–Meyer–Olkin (KMO) measure confirmed the suitability of the data for PCA implementation. The optimal number of principal components was determined by retaining the first 4 components, which collectively accounted for over 80% (precisely 82.74%) of the total variance.

Notably, the initial composite scores exhibited negative values. To enhance metric comparability, this paper adopts the standardization approach developed by Han et al. (2019), applying the linear transformation in Equation (7) to rescale the composite scores and construct the Digital Economy Comprehensive Development Index. Because the scores are divided by their range without first subtracting the minimum, the transformation does not map them onto a fixed interval; the resulting index ranges from 0.513 to 0.913 (Table 6). The PCA-based index is used as an alternative measure in the robustness test.

$$pca_i = \frac{g_i}{\max(g_i) - \min(g_i)} \cdot 0.4 + 0.6 \quad (7)$$

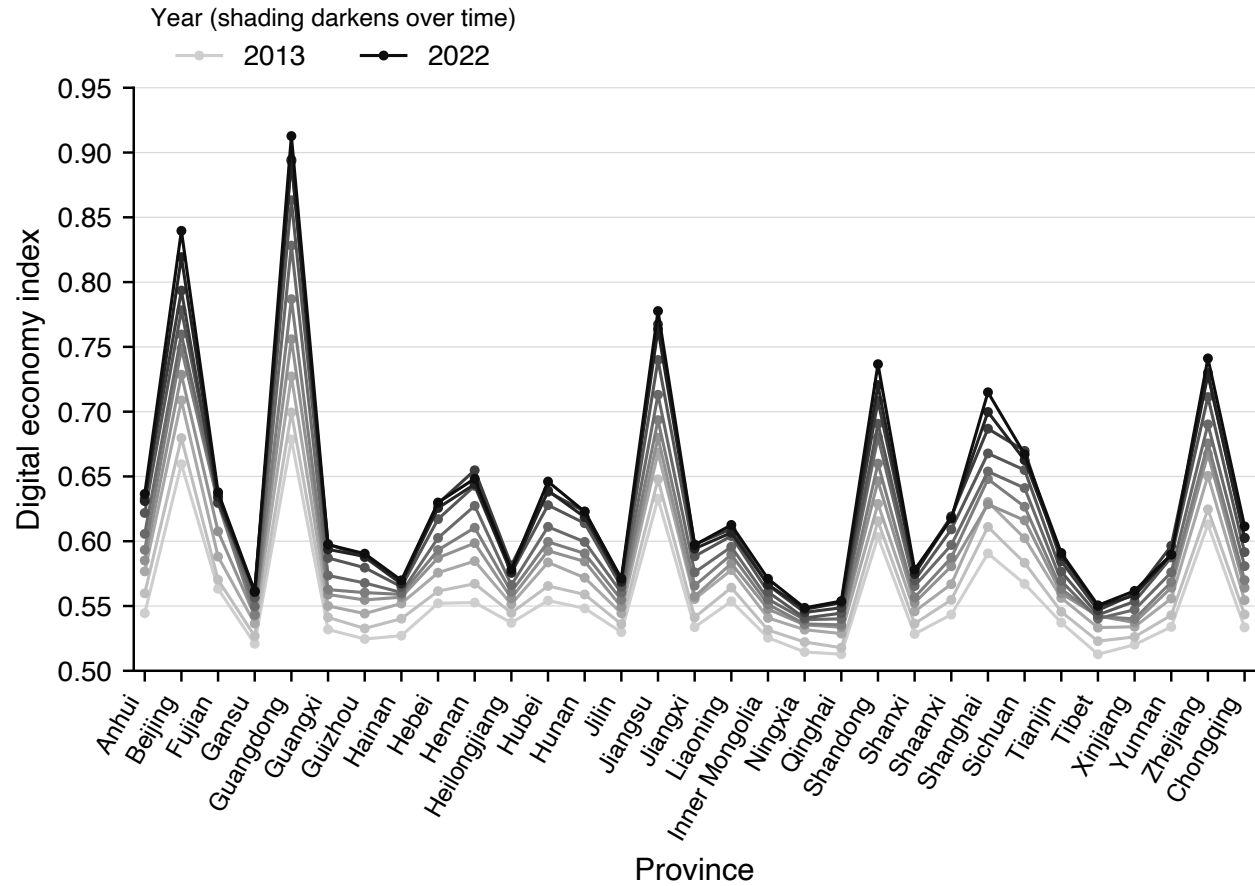
Note: g_i is the comprehensive score of province–year i ; $\max(g_i)$ and $\min(g_i)$ are the maximum and minimum comprehensive scores, respectively.

Figures 3 and 4 illustrate China's digital economy composite index as calculated by PCA.

The index of the development level of the digital economy obtained by the principal component analysis is highly consistent with the result calculated by the entropy weight method in terms of trend. Since the PCA-based index is not the core explanatory variable of the main

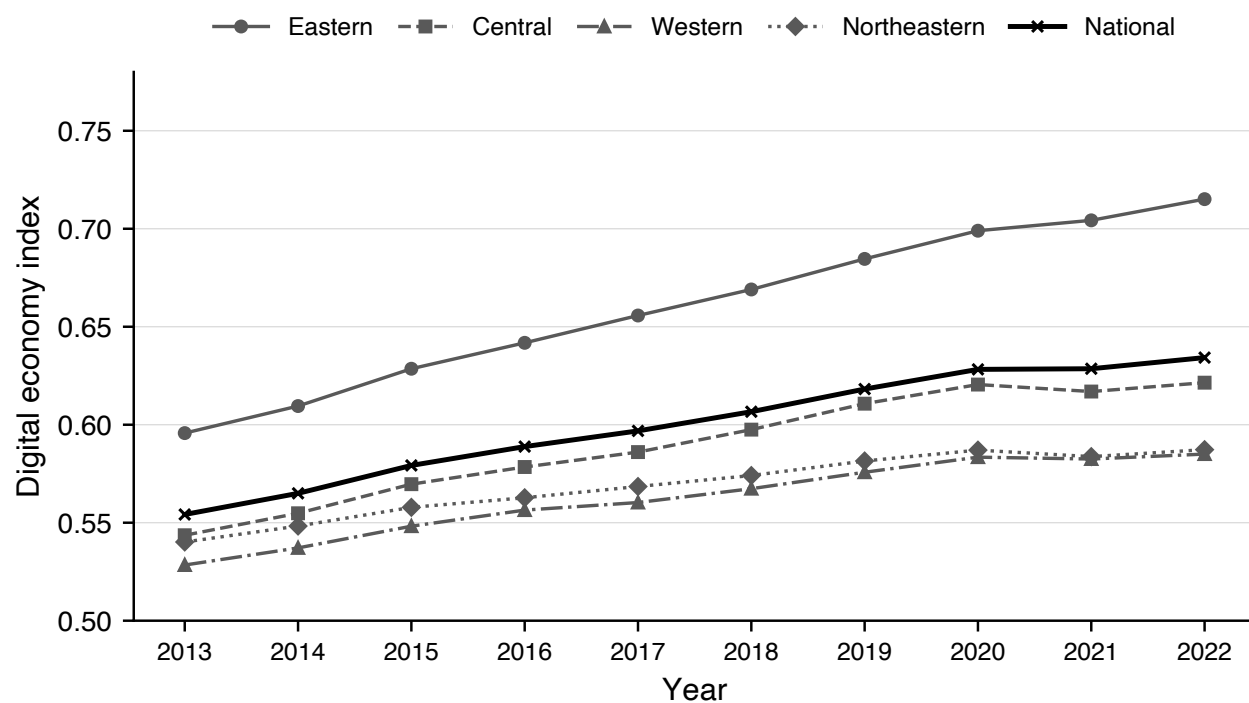
Figure 3

Digital Economy Comprehensive Development Index of 31 Provincial-Level Administrative Units (PCA)



Note. PCA = principal component analysis. Line shading darkens from 2013 to 2022.

regression, its values will not be described in detail here. However, its consistency with EWM ($r = .985$) provides a good basis for the subsequent robustness test. Table 5 shows the compound annual growth rates and the average digital economy development indices of the four regions.

Figure 4*Regional Digital Economy Development Index (PCA)*

Note. PCA = principal component analysis.

Table 5*CAGR and Average Index by Region (PCA)*

Statistic	Eastern	Western	Central	Northeastern	National
Index	0.660	0.562	0.590	0.569	0.600
CAGR	2.05%	1.14%	1.50%	0.93%	1.51%

Note. CAGR = compound annual growth rate. PCA = principal component analysis.

Data Sources and Descriptive Statistics

The data employed in this study were obtained from National Bureau of Statistics publications, including the China Statistical Yearbook, China Tertiary Industry Statistical Yearbook, China Education Statistical Yearbook, China Science and Technology Statistical Yearbook, and China Fixed Assets Investment Statistical Yearbook. Additional sources include provincial statistical communiqués on national economic and social development. Data on digital–real economy integration were sourced from Zhongli Information Network, while the digital inclusive finance index was derived from the Digital Finance Research Center of Peking University (Guo et al., 2020). The educational composition of employment used for the dependent variables comes from the China Labor Statistical Yearbook, and the number of students in higher education used for human capital from the China Science and Technology Statistical Yearbook and provincial statistical yearbooks. Missing values in the indicator series (90 of 6,820 cells: IPv4 addresses, 63; mobile internet users, 26; R&D institutions, 1) were filled by linear interpolation, or by linear extrapolation for the 27 cells in 2013. The graduate-degree share of Tibet, missing in the employment data for 2013 and 2014, was filled by linear extrapolation (−0.008% and 0.070%). The composite index is recomputed from the 22 indicators for every province–year; one value in the original data file (Tibet, 2018) did not match this computation (0.124 against 0.118), and the recomputed value is used. In the control-variable data file, the 2020 numbers of students in higher education, from which human capital is computed, are shifted by one row for the 16 provinces from Beijing to Henan in the order of province codes, so that each value is recorded under the next province (the value recorded for Beijing, for example, is Henan’s); the values are moved back, which aligns every province with its 2019–2021 trend and reproduces the figures in the 2020 education or statistical bulletins of Henan, Shandong, Guangdong, Hebei, Hubei, and Guizhou.

The variables used for the empirical analysis in this paper are based on data for 31 provincial-level administrative units in China (excluding the Hong Kong and Macao Special Administrative Regions and Taiwan Province), covering the period from 2013 to 2022. The descriptive statistics are shown in Table 6.

Table 6*Descriptive Statistics*

Variable	Obs	Mean	Std. Dev.	Min	Max
low_skill_pct	310	21.090	12.050	2.100	71.232
mid_skill_pct	310	65.906	10.566	22.159	83.175
high_skill_pct	310	10.605	7.429	2.581	46.908
digital_e	310	0.224	0.121	0.052	0.737
digital_e2	310	0.065	0.080	0.003	0.543
digital_p	310	0.600	0.071	0.513	0.913
digital_p2	310	0.365	0.094	0.263	0.833
human_capital	310	0.021	0.006	0.009	0.044
rd_intensity	310	0.018	0.012	0.002	0.068
digitalization	310	0.065	0.057	0.015	0.290
gov_intervention	310	0.290	0.203	0.104	1.354
industry	310	0.314	0.085	0.070	0.510

Note. Obs = number of province–year observations. digital_e2 and digital_p2 are squares of the uncentered indices; the regressions in Tables 7–9 center each index at its sample mean before squaring.

Empirical Results

Benchmark Regression Analysis

The regression results in column (2) of Table 7 reveal that both the linear and the squared terms of the centered index are statistically significant at the 5% level, with a negative squared term, indicating a nonlinear inverted U-shaped relationship between digital economy development and low-skilled employment. This dynamic is characterized by an initial phase of employment promotion followed by a subsequent substitution effect that, within the observed data, weakens the positive association without yet turning it significantly negative: In the early stages of digital economy development, the expansion of e-commerce, logistics networks, and basic digital infrastructure creates new low-skilled positions such as warehouse workers, delivery personnel, and entry-level data clerks, which directly absorb labor from traditional sectors (e.g., brick-and-mortar retail and offline service industries; B. Wu & Yang, 2022). However, as digital technologies like artificial intelligence (AI), robotic process automation (RPA), and smart

manufacturing systems mature, these technologies increasingly replace routine and repetitive tasks historically performed by low-skilled workers. For example, automated sorting systems in logistics reduce the need for manual package handling, while AI-driven chatbots substitute for basic customer service roles. Such technological substitution would eventually lead to a decline in the proportion of low-skilled employment, consistent with the theoretical framework proposed by Wen et al. (2019), which emphasizes the dual effects of digitalization on labor markets through task reallocation. The empirical estimation results of the model are reported in Table 7.

Columns (3) and (4) of Table 7 show that digital economy development has no significant association with the proportion of medium-skilled employment. This null association can be attributed to the simultaneous presence of substitution and creation effects within this skill category. On one hand, digitalization disrupts medium-skilled jobs involving standardized processes, such as routine administrative tasks, basic accounting, and production line quality control, which are increasingly automated through software systems or robotic technologies. On the other hand, new medium-skilled roles emerge to support digital infrastructure and technology adoption, including maintenance technicians, junior data analysts, and digital platform operators. These opposing forces, the destruction of old roles and the creation of new ones, offset each other, resulting in no significant net change in the share of medium-skilled employment in the context of the digital economy. As noted by Y. Zhao and Said (2023), this balance reflects the transitional nature of medium-skilled labor, which occupies a middle ground between the routine tasks vulnerable to automation and the complex cognitive tasks requiring high-skilled expertise.

In column (6), the linear term of the centered index is significantly negative at the 5% level, while the squared term is significantly positive at the same level, indicating a U-shaped relationship with high-skilled employment. The underlying mechanism unfolds in two stages: during the early phase of digital technology upgrading, traditional high-skilled positions tied to pre-digital industrial structures may decline as organizations restructure to adopt digital tools, leading to a temporary reduction in high-skilled employment shares. However, as digital technologies become deeply integrated into production and service systems, the demand for

Table 7*Benchmark Regression for the Three Skill Groups*

Variable	(1) low_skill_pct	(2) low_skill_pct	(3) mid_skill_pct	(4) mid_skill_pct	(5) high_skill_pct	(6) high_skill_pct
digital_e (centered)	12.237 (8.065)	34.442** (13.705)	−2.928 (7.540)	−13.911 (13.573)	−2.941 (4.431)	−16.550** (7.233)
digital_e ² (centered)		−34.308** (14.395)		16.970 (13.592)		21.027** (9.243)
human_capital	26.450 (133.063)	−52.694 (121.415)	366.488** (171.694)	405.635** (167.787)	−372.960** (136.687)	−324.453** (149.895)
rd_intensity	−155.801* (82.796)	−169.193** (78.372)	145.424* (85.323)	152.048* (85.727)	−41.004 (128.759)	−32.797 (122.520)
digitalization	−10.505 (7.725)	−9.703 (7.482)	7.047 (8.749)	6.651 (8.671)	−3.739 (3.557)	−4.230 (3.675)
gov_intervention	10.321 (8.445)	11.121 (7.611)	−2.110 (7.889)	−2.506 (7.519)	−5.134 (4.145)	−5.625 (4.607)
industry	7.520 (10.345)	10.998 (10.150)	−2.905 (10.291)	−4.625 (10.718)	−4.997 (6.801)	−7.129 (6.795)
Constant	19.738*** (6.817)	22.051*** (6.159)	62.973*** (6.420)	61.829*** (6.162)	17.730*** (5.477)	16.313*** (5.380)
Province and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	310	310	310	310	310	310
Adjusted <i>R</i> ²	.301	.320	.734	.734	.804	.809
Turning point		0.726 (0.075)		—		0.617 (0.106)
Lind–Mehlum test <i>p</i>		.443		.272		.188
Wild bootstrap <i>p</i> , squared term		.063		.283		.065

Note. All models include province and year fixed effects. Standard errors clustered by province in parentheses. digital_e is centered at its sample mean (0.224) before squaring, so the linear coefficient in columns (2), (4), and (6) is the marginal association at the mean. The turning point is expressed in the original units of digital_e, with its delta-method standard error in parentheses; it is not reported for column (4), where the squared term is not significant. The Lind–Mehlum test examines whether the slope changes sign within the observed range of digital_e (0.052–0.737; Lind & Mehlum, 2010). Wild cluster bootstrap *p* values use 9,999 Rademacher replications clustered by province (Roodman et al., 2019). **p* < .10. ***p* < .05. ****p* < .01.

advanced skills rises. New high-skilled roles emerge in specialized domains like algorithm development, big data analytics, AI system design, and digital strategy consulting, which require advanced technical and analytical capabilities. For instance, the proliferation of fintech and smart manufacturing has spurred hiring for data scientists and cybersecurity experts, which may drive a rebound in high-skilled employment. This upward segment, observed only for Guangdong in 2018–2022 and not confirmed by the Lind–Mehlum test, aligns with Jing and Sun’s (2019) observation that digital transformation ultimately increases demand for high-skilled labor after an initial adjustment period, as the economy shifts toward knowledge-intensive activities that rely on complex problem-solving and technological innovation.

Three qualifications apply to the shape of these relationships. First, the turning point of the low-skilled curve is 0.726 (95% CI [0.579, 0.873]), just below the largest observed value of the index (0.737, Guangdong in 2022). Only that province–year lies beyond it, the estimated slope at the sample maximum is close to zero (-0.76 , $SE = 5.24$), and the Lind–Mehlum test does not confirm a turning point within the data ($p = .443$). The downward segment of the inverted U is therefore a projection; what the data show directly is a positive association that weakens as the digital economy develops. Second, the turning point of the high-skilled curve is 0.617 (95% CI [0.409, 0.826]), inside the observed range, but only 5 of 310 province–years, all of them Guangdong in 2018–2022, lie beyond it, the slope at the sample maximum is not significant (5.03, $SE = 5.58$), and the Lind–Mehlum test does not confirm the U-shape ($p = .188$); what the data show directly is a negative association that weakens as the digital economy develops. Third, because there are only 31 clusters, the squared terms were re-tested with the wild cluster bootstrap, under which both are significant only at the 10% level ($p = .063$ for low-skilled and $.065$ for high-skilled employment). In terms of magnitude, around the sample mean (from 0.174 to 0.274) an increase of 0.10 in the index is associated with a 3.44-percentage-point higher low-skilled share and a 1.65-percentage-point lower high-skilled share.

Robustness Tests

To avoid the influence of the choice of proxy variable on the regression results, this paper first adopts the alternative variable method, replacing the digital economic development level calculated by the entropy weight method (EWM) with that calculated by principal component analysis (PCA) and re-estimating the model (Table 8). In the quadratic models (columns 2, 4, and 6), the signs of the linear and squared terms are the same as in the benchmark for all three skill groups, but the linear and squared terms for low- and high-skilled employment are significant only at the 10% level. Because the two indices are almost perfectly correlated ($r = .985$), this check mainly shows that the results do not depend on the choice between the two weighting methods; it does not provide independent variation.

Table 8

Alternative Variable Method for the Robustness Test

Variable	(1) low_skill_pct	(2) low_skill_pct	(3) mid_skill_pct	(4) mid_skill_pct	(5) high_skill_pct	(6) high_skill_pct
digital_p (centered)	14.418 (11.498)	50.165* (25.987)	-5.626 (10.991)	-23.530 (25.137)	3.276 (6.882)	-26.691* (15.629)
digital_p ² (centered)		-87.174* (47.440)		43.662 (49.823)		73.078* (39.761)
human_capital	42.528 (141.918)	7.129 (129.481)	354.192** (172.221)	371.921** (169.265)	-350.655** (134.250)	-320.980** (144.860)
rd_intensity	-146.008* (83.982)	-153.150* (82.129)	148.517* (83.194)	152.094* (86.023)	-60.194 (129.509)	-54.207 (122.554)
digitalization	-10.928 (7.820)	-9.858 (7.547)	7.003 (8.722)	6.467 (8.664)	-3.186 (3.429)	-4.083 (3.583)
gov_intervention	10.419 (8.711)	10.741 (8.244)	-1.979 (7.900)	-2.140 (7.732)	-5.635 (4.153)	-5.905 (4.379)
industry	6.383 (10.297)	10.093 (10.450)	-2.942 (10.077)	-4.800 (10.756)	-3.767 (6.736)	-6.877 (6.281)
Constant	19.385*** (6.955)	20.657*** (6.616)	63.096*** (6.450)	62.459*** (6.371)	17.695*** (5.378)	16.629*** (5.364)
Province and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	310	310	310	310	310	310
Adjusted <i>R</i> ²	.295	.308	.734	.734	.804	.811

Note. All models include province and year fixed effects. Standard errors clustered by province in parentheses. digital_p is centered at its sample mean (0.600) before squaring. * $p < .10$. ** $p < .05$.

*** $p < .01$.

Table 9 reports further checks on the quadratic specification. Rows (2)–(4) remove from the index the indicators that are themselves labor-market or education stocks (the share of IT-related employment, bachelor’s degree graduates, and R&D personnel), the indicator that enters the instrumental variable in the next section (mobile internet users), and all four together. The inverted U-shape for low-skilled employment and the U-shape for high-skilled employment remain significant with similar magnitudes, so the results are not driven by indicators that overlap with the dependent variables. Row (5) uses equal indicator weights and gives the same conclusions, as expected given that the entropy weights used in this paper are close to uniform. Row (6), however, shows that with the conventional entropy constant $k = 1/\ln(n)$ neither nonlinear relationship is statistically significant ($p = .186$ and $.197$ for the squared terms); the benchmark results therefore depend on a near-equal weighting of the 22 indicators. Row (7) excludes Guangdong, the province with the highest index values: the squared term for high-skilled employment is then no longer significant ($p = .125$), although its point estimate is larger, and the squared term for low-skilled employment is significant only at the 10% level ($p = .090$), while its linear term remains positive and significant. Row (8) shows that adding the 2015–2018 shortfall of the three shares to the medium-skilled share leaves the null result for medium-skilled employment unchanged; the slope of the shortfall with respect to the index, evaluated at the sample mean, is not significant ($b = -3.98$, $SE = 6.39$). Finally, because the three shares form a composition, the model was also estimated with log-ratio outcomes relative to medium-skilled employment: the squared term is significantly negative for $\ln(\text{low}/\text{mid})$ ($b = -2.53$, $p = .012$) but not significant for $\ln(\text{high}/\text{mid})$ ($b = 1.07$, $p = .352$). Taken together, both nonlinear associations are robust to the composition of the index and to equal weighting, but not to the conventional entropy constant; in addition, the curvature of the high-skilled relationship does not survive the exclusion of Guangdong or the log-ratio specification, and that of the low-skilled relationship weakens to the 10% level without Guangdong.

Table 9*Further Robustness Checks for the Quadratic Specification*

Specification	low_skill_pct		mid_skill_pct		high_skill_pct	
	Linear	Squared	Linear	Squared	Linear	Squared
(1) Baseline (Table 7)	34.442** (13.705)	-34.308** (14.395)	-13.911 (13.573)	16.970 (13.592)	-16.550** (7.233)	21.027** (9.243)
(2) Without labor-market indicators	31.774** (12.532)	-31.724** (13.199)	-12.545 (13.078)	15.347 (12.763)	-15.398** (6.822)	19.289** (8.466)
(3) Without mobile internet users	35.034** (13.867)	-35.937** (15.281)	-14.086 (13.668)	16.906 (14.515)	-16.353** (7.359)	22.726** (10.133)
(4) Without (2) and (3) combined	31.973** (12.612)	-33.022** (14.066)	-12.570 (13.091)	15.037 (13.664)	-14.941** (6.862)	20.889** (9.308)
(5) Equal indicator weights	33.129** (13.828)	-33.419** (14.509)	-14.606 (13.528)	18.055 (13.655)	-15.765** (7.204)	20.431** (9.189)
(6) Entropy constant $k = 1/\ln(n)$	18.054 (13.176)	-18.068 (13.336)	-14.086 (12.310)	17.626 (12.936)	-7.901 (7.721)	11.831 (8.960)
(7) Excluding Guangdong	38.014** (15.493)	-46.757* (26.672)	-12.108 (14.747)	10.214 (28.649)	-19.970** (8.563)	36.929 (23.402)
(8) Mid share adjusted, 2015–2018			-17.878 (12.473)	13.306 (13.475)		

Note. Each row re-estimates columns (2), (4), and (6) of Table 7, changing only the element named in the row label; the index is centered at its own full-sample mean before squaring. The labor-market indicators are the percentage of IT-related employment, the number of bachelor's degree graduates, and the full-time equivalent of R&D personnel (Table 2). Row (6) recomputes the entropy weights with $k = 1/\ln(310)$ instead of $1/\ln(10)$. Row (7) has $N = 300$; all other rows have $N = 310$. Row (8) adds the 2015–2018 shortfall of the three shares from 100 to mid_skill_pct and therefore affects only the medium-skilled equation. All models include the control variables and province and year fixed effects. Standard errors clustered by province in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$.

Endogeneity Test

Although the benchmark regression model has controlled for some variables that affect the distribution of human capital, due to data availability issues for some unobservable variables, the empirical analysis may still suffer from endogeneity bias, thereby affecting the validity of causal inference about the impact of the digital economy on the distribution of human capital. For this reason, this paper adopts the instrumental variable method to identify and handle the endogeneity problem of the core explanatory variable.

In the process of selecting instrumental variables, this paper follows Huang et al. (2019), who interact a historically predetermined variable with a time-varying one, and Ye et al. (2021), and combines a geographical variable with a time-varying one: the reciprocal of provincial terrain undulation, which raises the cost of building digital infrastructure, is multiplied by the number of mobile internet users, and the logarithm of the product (*ivv*) is used as the instrument for the digital economy index. Table 10 reports the results. The first-stage coefficient of *ivv* is positive and significant at the 1% level, the Kleibergen–Paap rk LM statistic rejects under-identification at the 1% level, and the Cragg–Donald Wald F statistic (203.76) exceeds the Stock–Yogo 10% critical value of 16.38. Since the number of instrumental variables is exactly equal to the number of endogenous variables, the model is exactly identified, and no over-identification test (Hansen test) is available.

These statistics nevertheless do not establish that the instrument is valid, for two reasons. First, because terrain undulation does not vary over time, its logarithm is absorbed by the province fixed effects, so the within-province variation of *ivv* is identical to that of the logarithm of mobile internet users: using that logarithm alone as the instrument reproduces the second-stage estimates (for example, 57.605 for *digital_e* in column 4, against 57.605 with *ivv*). The geographical component therefore contributes no identifying variation, and when the product is used in levels, so that terrain does contribute, the instrument is weak (Cragg–Donald Wald F = 7.87). Second, the number of mobile internet users is one of the 22 indicators of the digital economy index (Table 2), so the strong first stage partly reflects a mechanical relationship, and

Table 10*Endogeneity Test by the Two-Stage Least Squares Method*

Variable	(1) First stage: digital_e	(2) low_skill_pct	(3) mid_skill_pct	(4) high_skill_pct
ivv	0.068*** (0.005)			
digital_e		−34.326** (15.613)	−201.691*** (24.561)	57.605*** (11.941)
digital_e2	0.859*** (0.042)	34.095** (14.872)	169.153*** (25.826)	−33.659*** (11.090)
human_capital	1.070*** (0.319)	208.663*** (77.728)	842.888*** (124.251)	80.538 (62.583)
rd_intensity	1.168*** (0.265)	−53.952 (48.027)	362.408*** (124.461)	12.648 (72.690)
digitalization	0.018* (0.010)	−9.159*** (2.733)	33.981*** (4.185)	−0.763 (1.529)
gov_intervention	0.017 (0.023)	14.607* (8.002)	−24.962*** (7.482)	−14.114*** (2.983)
industry	−0.147*** (0.030)	4.497 (9.584)	−19.154 (11.715)	3.006 (6.312)
Kleibergen–Paap rk LM statistic		52.175	52.175	52.175
$\chi^2(1)$ <i>p</i> value		< .001	< .001	< .001
Cragg–Donald Wald <i>F</i> statistic		203.764	203.764	203.764
Kleibergen–Paap rk Wald <i>F</i> statistic		184.689	184.689	184.689
Province fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	No	No	No	No
<i>N</i>	310	310	310	310
Adjusted <i>R</i> ²	—	.041	.240	.619

Note. The excluded instrument is $ivv = \ln[(1/\text{terrain undulation}) \times \text{mobile internet users}]$; digital_e2 and the control variables are included instruments. digital_e and digital_e2 are not centered (unlike in Tables 7–9), so the digital_e coefficient is the slope at an index value of zero. Column (1) is the first stage shared by columns (2)–(4). All columns include province fixed effects but no year fixed effects. Robust standard errors in parentheses. Because terrain undulation does not vary over time, its logarithm is absorbed by the province fixed effects, and the within-province variation of ivv is identical to that of $\ln(\text{mobile internet users})$, which is itself an indicator of the digital economy index (Table 2). The estimates are therefore not interpreted as causal. * $p < .10$. ** $p < .05$. *** $p < .01$.

the exclusion restriction, namely that internet diffusion affects the skill composition of employment only through the index, is difficult to defend. The second-stage estimates, which, unlike the benchmark, omit year fixed effects, also depart from the benchmark: the low-skilled curvature is reversed (a U rather than an inverted U; $p = .028$ and $.022$ for the linear and squared terms), the medium-skilled terms are highly significant although the benchmark finds no significant association, and the high-skilled curvature is also reversed (concave rather than convex, although the implied maximum, at 0.856, lies beyond the largest observed index value of 0.737); removing mobile internet users from the index gives similar low-skilled estimates. The instrumental-variable results are therefore reported for completeness, and the benchmark estimates in this paper are interpreted as conditional associations rather than causal effects. A historically predetermined instrument that is unrelated to the index indicators, of the kind used by Huang et al. (2019), would provide a more credible basis for causal inference.

Heterogeneity

Drawing on the literature, which emphasizes that regional digital economy development is strongly correlated with geographical location and economic endowments, this study classifies the 31 provincial-level administrative units into inland and coastal provinces based on geographical proximity to the sea (20 inland and 11 coastal provinces; Hebei, which borders the Bohai Sea, is coded as coastal), conducting a sub-sample regression analysis, in which the index enters linearly (without its square), to explore heterogeneous regional associations. The results, presented in Table 11, reveal distinct associations across these two groups:

For inland provinces, the regression coefficient of the digital economy development level on low-skilled employment is positive and significant at the 5% level, indicating a strong positive association. This can be attributed to their position in the early stages of digitalization, where technological applications are predominantly oriented toward labor-intensive settings rather than capital- or technology-intensive ones. Specifically, the rapid expansion of e-commerce platforms and on-demand service industries in inland regions has spurred demand for low-skilled jobs in logistics (e.g., warehouse sorting, delivery personnel for Alibaba's Cainiao Network), food

Table 11*Regional Heterogeneity Analysis*

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Inland provinces			Coastal provinces		
	Low	Mid	High	Low	Mid	High
digital_e	43.341** (16.044)	−33.526*** (8.533)	−9.569 (8.007)	1.929 (5.062)	10.498 (11.360)	−5.479 (8.076)
human_capital	56.337 (180.027)	354.222* (186.790)	−353.741** (128.904)	−70.313 (138.016)	354.872 (257.567)	−403.825** (175.683)
rd_intensity	−380.994** (164.611)	314.650** (136.903)	199.535** (81.281)	−114.206 (77.748)	62.237 (150.314)	−27.865 (118.580)
digitalization	−4.288 (10.354)	2.302 (10.671)	−3.576 (3.671)	−18.719** (5.983)	11.417 (7.449)	9.302 (8.662)
gov_intervention	8.161 (10.582)	−4.715 (8.461)	1.039 (2.770)	17.496* (9.344)	5.298 (19.209)	−37.568** (15.858)
industry	8.515 (14.117)	−6.301 (14.842)	4.506 (7.657)	−4.430 (13.552)	30.829 (37.488)	−10.212 (27.095)
Constant	18.926* (10.673)	65.169*** (10.422)	9.346* (4.482)	18.094** (6.038)	52.847*** (16.130)	28.488** (11.692)
Province and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	200	200	200	110	110	110
Adjusted <i>R</i> ²	.316	.719	.850	.443	.816	.801

Note. Low, Mid, and High denote low_skill_pct, mid_skill_pct, and high_skill_pct. The inland sample contains 20 provinces and the coastal sample 11; Hebei, which borders the Bohai Sea, is coded as coastal. The models are linear in digital_e and include province and year fixed effects. Standard errors clustered by province in parentheses. With only 11 clusters and 15 slope parameters, the cluster-robust standard errors in columns (4)–(6) are unreliable; Table 12 reports a pooled test. * $p < .10$. ** $p < .05$. *** $p < .01$.

delivery services (e.g., Meituan and Ele.me riders), and basic digital service support (e.g., data annotation clerks for AI training). These sectors rely heavily on human labor due to less advanced automation adoption than in coastal areas, creating a short-term surge in low-skilled employment. Conversely, the coefficient for medium-skilled employment is significantly negative at the 1% level, suggesting a negative association. This arises from digital technologies such as automated production lines, intelligent inventory management systems, and robotic process automation (RPA) replacing routine medium-skilled roles in traditional manufacturing and services. For example, in Hubei Province's automotive industry, the introduction of robotic assembly lines in Dongfeng Motor's factories has reduced demand for medium-skilled technicians specializing in manual quality control and process supervision, illustrating how automation displaces standardized technical positions in inland industrial hubs (X. Zhao, 2017). The regression coefficient for high-skilled employment is statistically insignificant, reflecting a lack of robust demand for advanced roles. This gap is likely due to the limited technological R&D capabilities and innovation ecosystems of most inland provinces (the inland group nevertheless includes research-intensive Beijing); without strong local tech enterprises or research institutions, the need for high-skilled jobs in AI development, big data analytics, or digital strategy remains latent. Regarding the control variables, R&D intensity is significantly negatively associated with low-skilled employment but positively associated with medium-skilled labor. This dual association suggests that increased R&D drives technological substitution: automated equipment reduces reliance on low-end manual labor, while simultaneously expanding tech-intensive industries that require skilled workers for equipment maintenance, software debugging, and process optimization, thereby creating new medium-skilled positions.

In coastal provinces, the coefficients of the digital economy index are statistically insignificant for low-, medium-, and high-skilled employment alike. The coastal sub-sample, however, contains only 11 provinces, and with 15 slope parameters its cluster-robust standard errors are unreliable (wild cluster bootstrap p values with Webb weights are .701, .431, and .631). To test whether the association actually differs between the two groups, the model was

re-estimated on the full sample with an interaction between the index and a coastal indicator (Table 12). The positive association between the digital economy and low-skilled employment is significantly weaker in coastal provinces: the coefficient is 25.88 in inland provinces and 14.20 lower in coastal provinces ($p = .016$; wild cluster bootstrap $p = .022$). No significant regional difference is found for medium- or high-skilled employment in this pooled model, which constrains the coefficients of the control variables and year effects to be equal in both groups. Coding Hebei as inland, as in the original data file, gives a similar interaction for low-skilled employment ($b = -12.35$, $p = .039$). This pattern aligns with the advanced digital economy in coastal areas. Specifically, the Yangtze and Pearl River Deltas are dominated by tech clusters that have moved away from labor-intensive work. However, the data do not show that digital development is associated with higher high-skilled employment in these coastal provinces. Among the control variables, government intervention is positively associated with low-skilled employment ($p = .091$) and negatively associated with high-skilled employment ($p = .039$) in coastal areas, although these p values rest on the unreliable 11-cluster standard errors. These signs are the opposite of what an explanation based on subsidies for automation and industrial upgrading policies such as “Made in China 2025” would predict, and given the small coastal sample they should be interpreted with caution.

Taken together, these regional disparities highlight the importance of tailoring digital policies to local development stages: inland areas require strategies to balance low-skilled job creation with gradual skill upgrading, while coastal regions, where the low-skilled association is already weaker, need to prepare for skill upgrading and foster innovation spillover to inland provinces.

Table 12*Regional Heterogeneity: Interaction Test*

Variable	(1) low_skill_pct	(2) mid_skill_pct	(3) high_skill_pct
digital_e (inland provinces)	25.885** (9.418)	-11.769 (12.501)	-7.461 (9.067)
digital_e × Coastal	-14.202** (5.537)	9.200 (8.089)	4.704 (7.372)
Implied slope in coastal provinces	11.683* (6.747)	-2.569 (6.438)	-2.757 (4.229)
Wild bootstrap p , interaction	.022	.325	.586
Control variables	Yes	Yes	Yes
Province and year fixed effects	Yes	Yes	Yes
N	310	310	310
Adjusted R^2	.321	.736	.805

Note. Full-sample regressions with an interaction between digital_e and an indicator for the 11 coastal provinces; the coastal main effect is absorbed by the province fixed effects. The implied slope in coastal provinces is the sum of the first two coefficients. All models include the control variables of Table 11 and province and year fixed effects, with coefficients common to both groups (hence the inland coefficients differ from those in Table 11). Standard errors clustered by province (31 clusters) in parentheses; wild cluster bootstrap p values use 9,999 Rademacher replications (Roodman et al., 2019). * $p < .10$. ** $p < .05$. *** $p < .01$.

Policy Implications

This paper finds that the digital economy has a U-shaped association with high-skilled employment and an inverted U-shaped association with low-skilled employment, although neither turning point is confirmed by the Lind–Mehlum test and the upward segment of the high-skilled relationship is observed only for Guangdong. Moreover, the positive effect of digital skills training on high-skilled employment is 2.3 times that of traditional training, and it is necessary to strengthen institutional support for “industry-education integration” (Jia, 2023). Based on this, this paper offers the following suggestions: Establish a lifelong digital learning account, led by the government and funded jointly by enterprises and individuals. Allow workers to use the account balance to purchase online digital courses, which can help to build a framework for

enhancing the digital literacy of all citizens, narrow the educational gap among the employed labor force, and optimize the allocation of human resources.

The heterogeneity analysis in this paper shows that the positive association between the digital economy and low-skilled employment is significantly weaker in coastal provinces than in inland provinces, where digital development is still associated with a higher low-skilled employment share. Using a spatial Durbin model, the relevant literature finds that the employment effect of the digital economy in the central and western regions depends on the technology spillover from the east. It is suggested that “targeted support for the digital economy” be provided (Lu & Zhou, 2023). Based on this, it is proposed to build a “Digital Talent Port” in the eastern region, pilot a “Special Cooperation Zone for Digital Economy,” allow overseas digital talent to work visa-free with digital technology, and establish a mechanism for the return of overseas digital talent, modeled on Singapore’s “Global Alumni Network Program.” The focus should be on developing employment in cutting-edge fields such as blockchain and the metaverse. In the inland areas, “Eastern Digital Enterprise Collaboration Parks” will be established in hub cities such as Zhengzhou and Chengdu. Eastern enterprises will provide technology, while the governments of the central and western regions will offer venues and labor training subsidies. These parks will develop adaptable industries such as data annotation and intelligent hardware assembly and leverage the spillover effects of the eastern digital economy to improve employees’ education level in the central region.

Limitations and Future Research

Although this study revealed the macro-level associations between the digital economy and employment through a panel model, due to the limitations of data availability, it failed to capture the impact of breakthroughs in generative AI technology on content creation positions after 2023. Furthermore, the study did not incorporate the moderating effect of institutional variables (such as the progress of household registration reform), which might lead to an underestimation of the impact of policy synergy on the optimization of the regional employment structure. Several further limitations follow from the empirical design. First, the dependent

variables measure the educational composition of the employed population, which reflects the supply of educated workers as well as the demand for skills; occupation, task, or wage data would be needed to identify changes in skill demand. Second, the instrumental variable does not provide credible exogenous variation, so the estimates are conditional associations rather than causal effects. Third, neither turning point is confirmed by the Lind–Mehlum test: the low-skilled turning point lies at the edge of the observed data, and the upward segment of the high-skilled relationship is observed only for Guangdong, so the stage interpretation is partly a projection. Fourth, the results depend on the near-equal weighting of the index and are not robust to the conventional entropy constant; in addition, the high-skilled curvature is not significant without Guangdong or with log-ratio outcomes, and the low-skilled curvature is significant only at the 10% level without Guangdong. Fifth, the educational composition data are missing at least one category in 2015–2018, and the coastal sub-sample is too small for reliable cluster-robust inference. Future research can combine enterprise employment data with natural experiments (such as the establishment of the “National Digital Economy Innovation and Development Pilot Zone”) to identify causal effects.

Conclusion

The rapid advancement of China’s digital economy has been accompanied by profound transformations in the employment structure, characterized by distinct nonlinear dynamics and pronounced regional disparities. The empirical results suggest an inverted U-shaped relationship between the digital economy and low-skilled employment. Initially, e-commerce growth and 4G/5G network expansion create jobs in logistics, basic data entry, and digital infrastructure construction. Over time, automation and AI are expected to gradually replace these jobs. For instance, the rise of robotic process automation (RPA) in administrative tasks and smart storage systems in logistics may reduce demand for routine manual labor, reflecting a possible transition from employment promotion to displacement as digital technologies mature. However, in the observed data, the association weakens but does not become significantly negative. In contrast, high-skilled employment demonstrates a U-shaped trajectory: early-stage digitalization disrupts

traditional high-skilled roles tied to pre-digital industrial frameworks (e.g., conventional manufacturing engineering and analog financial analysis), but as digital integration deepens, especially in areas like cloud computing, big data analytics, and AI development, the demand for advanced positions such as algorithm engineers, cybersecurity specialists, and digital transformation consultants rises again, although this upward segment is observed only in the most digitally developed province-years and is not statistically confirmed. Medium-skilled employment remains relatively stable, possibly due to the offsetting effects of automation replacing standardized process jobs (e.g., routine quality control on production lines) and the emergence of new technical maintenance and platform operation roles that require moderate digital literacy, leading to no significant net change in its employment share.

Regionally, these associations differ between inland and coastal provinces. In inland provinces, digital development is still associated with a higher low-skilled employment share, plausibly in logistics (e.g., delivery services for e-commerce platforms) and basic digital services (e.g., customer support for online platforms), while medium-skilled employment is negatively associated with digital development in the sub-sample estimates (though not in the pooled interaction model), possibly through the automation of routine tasks in sectors like traditional retail and administrative services. In coastal provinces, the positive association with low-skilled employment is significantly weaker, but the data do not show a significant association between digital development and high-skilled employment there. This spatial divide highlights the unequal distribution of “digital dividends,” with inland areas facing challenges such as talent outflow, inadequate digital infrastructure, and structural vulnerabilities from rapid automation without commensurate skill upgrading.

The findings yield critical policy implications that emphasize the need for adaptive institutional frameworks to align digital advancement with inclusive employment growth. Key strategies include (a) establishing national lifelong learning systems focused on digital literacy and skills training, such as subsidized online courses in data analysis and AI basics, to bridge the growing skill gap between low- and medium-skilled workers and emerging digital roles; (b)

fostering cross-regional collaboration through initiatives like Digital Talent Exchange Hubs in eastern tech hubs, which facilitate talent mobility and knowledge transfer to inland regions via mentorship programs and joint R&D projects; and (c) creating Eastern-Inland Digital Industry Cooperation Parks in central and western cities, where eastern digital enterprises set up regional branches or innovation centers, channeling technology, capital, and management expertise into localized job creation—particularly in medium-skilled technical support and digital service sectors. However, the study acknowledges limitations, including the exclusion of post-2023 generative AI impacts (e.g., large language models transforming content creation and customer service roles), the absence of detailed institutional variables (e.g., regional digital policy effectiveness and labor market regulations), the use of educational attainment as a proxy for skills, the lack of a credible instrument for causal identification, and the sensitivity of the nonlinear results to the weighting of the index and, for high-skilled employment, to the exclusion of Guangdong. These gaps underscore the need for future research to incorporate firm-level microdata, natural experiments in digital economy pilot zones, and longitudinal analyses of skill-biased technological change. Ultimately, proactive policy design must balance the disruptive forces of digitalization with equitable investments in human capital and regional coordination, ensuring that China's digital transformation fosters sustainable, inclusive employment structures across all geographic and skill-based segments of the labor market.

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